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Enhanced carbon dioxide removal from coupled direct air capture–bioenergy systems†

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Engineered carbon dioxide (CO₂) removal technologies—such as direct air capture (DAC) and bioenergy with carbon capture and sequestration (BECCS)—require significant advancements to reduce costs and increase scale without environmental or social harm. Up until this point, advancements in BECCS and DAC technologies have progressed independently of each other, despite the ability of BECCS to provide carbon-negative heat and power to DAC processes. Here, we demonstrate synergistic integration of BECCS and DAC systems decreases costs, increases carbon removal, and extends the impact of scarce biomass resources. We estimate biomass integrated DAC systems can increase net CO₂ removal by 109–119% at lower costs than standalone reference systems. If hybrid BECCS–DAC systems were deployed across the US using biomass resources co-located with geologic storage, 1.5 Gt of CO₂ could be sequestered annually without long-distance biomass or CO₂ transport in 2030. Innovation in the design and implementation of coupled BECCS–DAC systems would have a profound impact on CO₂ removal, if realized.

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1. Introduction

Engineered CO₂ removal technologies—such as direct air capture (DAC) and bioenergy with carbon capture and sequestration (BECCS)—hold large promise for rapid and reliable CO₂ removal consistent with stringent climate change mitigation.^{1–3} However, DAC and BECCS remain commercially and technically immature.⁴ The Intergovernmental Panel on Climate Change (IPCC) insists that DAC and BECCS technologies must scale up rapidly as the necessity for CO₂ removal increases.³ To meet stringent climate targets set forth by the IPCC, the United States is expected to sequester 1 Gt-CO₂ per year in 2050 and 3 Gt-CO₂ per year in 2100.⁵ To meet such ambitious targets, research, development, and demonstration of CO₂ removal technologies must rapidly accelerate.^{3,6,7}

BECCS refers to the utilization of renewable biomass resources for energy and the subsequent capture and sequestration of CO₂ via injection into geologic formations.^{3,5,6} The advantages of using BECCS for engineered CO₂ removal include relatively low cost and the existence of established technologies for both biomass conversion and CO₂ capture and sequestration.⁶ The major drawbacks of BECCS technologies are their substantial land requirements, sustainability impacts at scale,

limited resource potential, and social barriers to acceptance.^{8–10} Nonetheless, the US National Academy of Sciences recently classified BECCS as one of four CO₂ removal technologies ready for immediate deployment.⁶ DAC refers to the direct removal of CO₂ from the atmosphere and subsequent sequestration. Two leading DAC technology platforms employ liquid solvents and solid sorbents.^{2,11} Several companies are currently commercializing technologies that use these platforms, including Carbon Engineering, which uses a liquid alkaline solvent for thermal energy-driven calcium looping,¹² and Global Thermostat and Climeworks, which use solid-supported amine sorbents for thermal energy-driven adsorption/desorption.^{13,14} The advantages of using DAC for engineered CO₂ removal include flexibility in site location, low land footprint, and potentially limitless scale. However, DAC technologies suffer from high costs, substantial energy requirements, and commercial immaturity.² The US National Academy of Sciences recently reported the levelized costs of CO₂ removal *via* DAC to be uneconomical in current policy environments.^{6,15}

The energy-intensive nature of liquid solvent and solid sorbent DAC technologies contribute to their high costs of operation. The majority of such DAC technologies under development require substantial quantities of thermal energy for regeneration (Fig. 1). Liquid solvent based calcium looping requires high temperature thermal energy to liberate CO₂ from calcium carbonate. A direct heating, oxygen-fired calciner using natural gas for fuel is utilized in Carbon Engineering's pilot plant in Squamish, BC, which they describe as their 'baseline' process in recent literature.¹² To ensure their process results in a net removal of CO₂, Carbon Engineering envisions the

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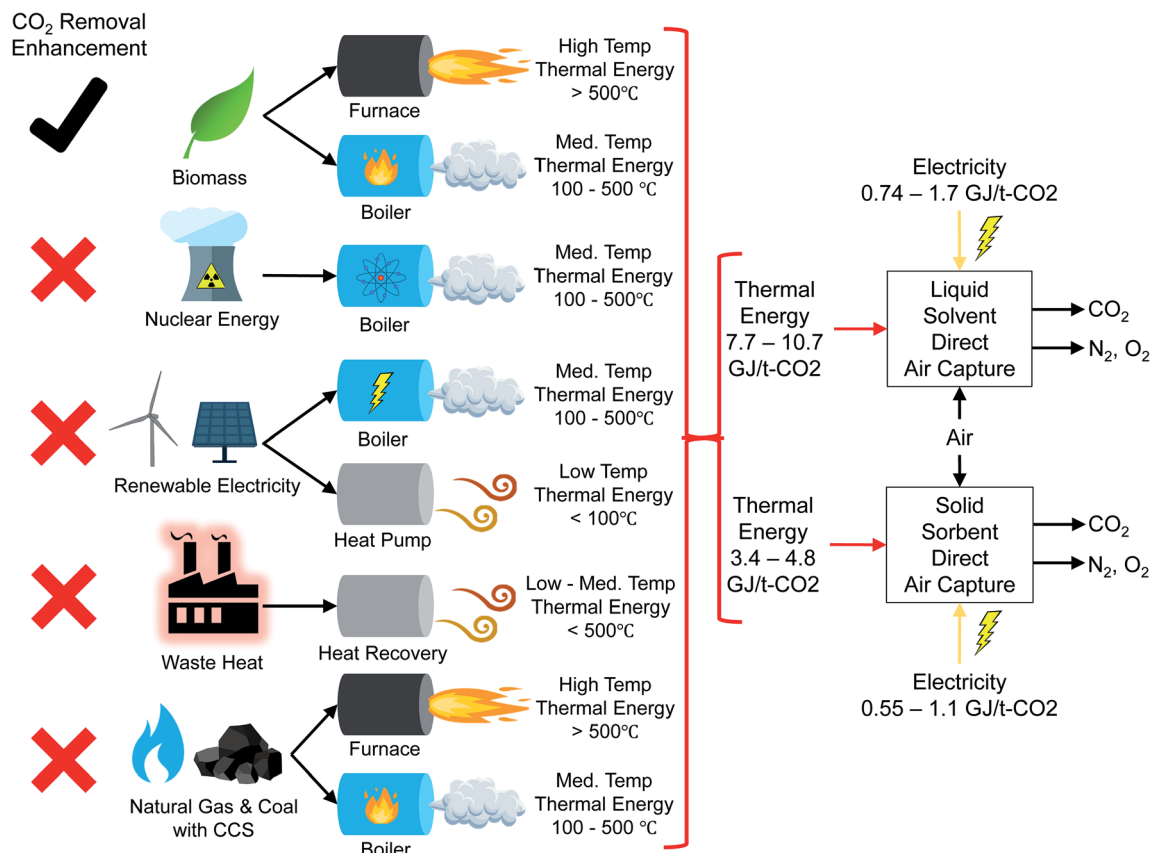


Fig. 1 Commercially established, low-carbon methods for obtaining thermal energy and their applicability towards carbon dioxide removal via direct air capture (left).¹⁷ Ranges of thermal and electrical energy demands for direct air capture recently reported by the US National Academy of Sciences (right) (values do not include energy required for CO₂ compression).⁶

sequestration of CO₂ produced from natural gas combustion along with that from the atmosphere. Similarly, solid-supported amine adsorption/desorption requires a significant quantity of low-to-medium temperature thermal energy for CO₂ liberation and sorbent regeneration.^{6,13,14} Carbon Engineering recently published a detailed techno-economic analysis of their pilot-scale liquid solvent process in which levelized costs were estimated to vary between \$126–170 per t-CO₂ for an *n*th-of-a-kind plant with a net removal of 0.98 Mt-CO₂ per year (12). The US National Academy of Sciences estimates costs of solid sorbent CO₂ removal to vary between \$89–877 per t-CO₂ for an *n*th-of-a-kind plant utilizing low temperature thermal energy with a net removal of ~1 Mt-CO₂ per year;⁶ the wide range of costs is due to the different sources of energy used in their modeling, namely solar, nuclear, natural gas, and coal (biomass was not included).

Many emerging DAC innovators claim that fossilized hydrocarbons, renewable electricity, or nuclear energy will provide the thermal energy necessary for operation.⁶ Biomass, however, is frequently neglected in such analyses: for instance, recent energy system modelling of DAC integration into low-carbon heat and power systems did not consider biomass as an energy source.¹⁶ Biomass combustion technologies hold several advantages over other low-carbon technologies for providing thermal energy for DAC, including high heating

efficiencies, high technology readiness, and the ability to contribute to CO₂ removal (Fig. 1).^{17,18}

The use of biomass for thermal energy-driven DAC technologies remains relatively unexplored, despite the ability of BECCS to provide carbon-negative heat and power to DAC processes. Here, we demonstrate that integration of BECCS and DAC systems reduces costs, increases CO₂ removal, and extends the impact of scarce biomass resources. To demonstrate this, we estimate the levelized costs of CO₂ removal and performance of two hybrid BECCS–DAC systems and compare them to commonly proposed standalone CO₂ removal systems.

2. Methodology

2.1 System design & modeling

We design, model, and analyze four CO₂ removal systems by scaling and integrating detailed technologies from published literature and using conservative assumptions to estimate performance and costs.^{6,12,19–21} The integration of technologies considered in this study demonstrate the benefits of synergizing bioenergy and direct air capture systems relative to standalone reference systems. We compare a natural gas plus liquid solvent direct air capture (NG + LS-DAC, Fig. 2A) system to a biomass gasification plus liquid solvent direct air capture (BG + LS-DAC, Fig. 2B), mimicking a biomass-enhanced DAC system.

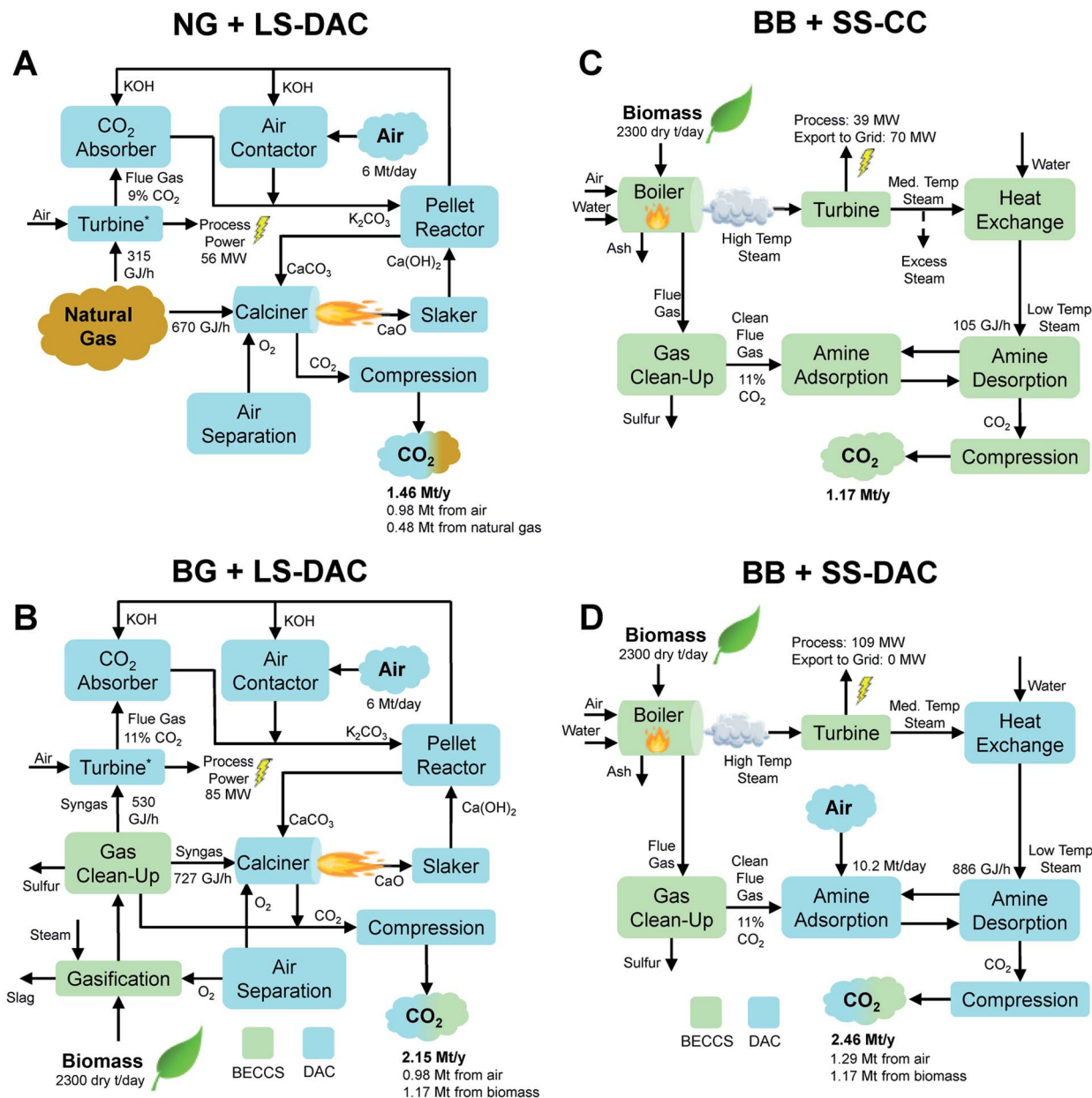


Fig. 2 Simplified process flow diagrams of the four baseline CO₂ removal systems. CO₂ concentrations based on volume. (A) Natural gas plus liquid solvent direct air capture (NG + LS-DAC) system, adapted from Carbon Engineering's baseline DAC system.¹² Turbine* includes a gas turbogenerator and a heat recovery steam generator. (B) Biomass gasification plus liquid solvent direct air capture (BG + LS-DAC) system.^{6,19} (C) Biomass boiler plus solid sorbent carbon capture (BB + SS-CC) system.^{6,20} (D) Biomass boiler plus solid sorbent direct air capture (BB + SS-DAC) system.^{6,20}

Subsequently, we compare a biomass boiler plus solid sorbent carbon capture (BB + SS-CC, Fig. 2C) system to a biomass boiler plus solid sorbent direct air capture (BB + SS-DAC, Fig. 2D) system, mimicking a DAC-enhanced BECCS system.

We design our biomass-DAC systems to demonstrate the inherent advantages biomass has as an energy source for DAC. Each system is modeled as a greenfield plant. We define system boundaries as inside the battery limits of the chemical processing facilities, with compressed CO₂ as the final product. Table S1† provides a basic overview of the four CO₂ removal systems. Fig. S1† shows the generalized approach to designing,

modeling, and analyzing the four systems. Table 1 provides pertinent technical information about the four CO₂ removal systems.

Eqn (1) is used to calculate the minimum work required for separation of CO₂ from various gaseous streams.²¹

Minimum work for CO₂ separation = $w_{\min}$

$$= RT \left[\begin{aligned} & (n_{r,\text{CO}_2} \ln(X_{r,\text{CO}_2}) + n_r \ln(X_r)) \\ & + (n_{p,\text{CO}_2} \ln(X_{p,\text{CO}_2}) + n_p \ln(X_p)) \\ & - (n_{i,\text{CO}_2} \ln(X_{i,\text{CO}_2}) + n_i \ln(X_i)) \end{aligned} \right] \quad (1)$$

Table 1 Technical details for the four baseline CO₂ removal systems analyzed in this study. The parameters for the natural gas plus liquid solvent (NG + LS-DAC) system were taken from Keith *et al.*'s technical analysis of Carbon Engineering's baseline direct air capture technology¹²

	NG + LS-DAC	BG + LS-DAC	BB + SS-CC	BB + SS-DAC
Air flow rate (Mt per day)	6.0	6.0	—	10.2
Biomass flow rate (dry t per day)	—	2300	2300	2300
Air-derived CO ₂ capture (Mt-CO ₂ per year)	0.98	0.98	—	1.29
Air-derived CO ₂ capture efficiency (t-CO ₂ removed/t-CO ₂ in air)	74.5%	74.5%	—	65%
Natural-gas derived CO ₂ captured (Mt-CO ₂ per year)	0.47	—	—	—
Biomass-derived CO ₂ captured (Mt-CO ₂ per year)	—	1.17	1.17	1.17
Flue gas CO ₂ capture efficiency (t-CO ₂ removed/t-CO ₂ generated)	90% nat. gas	90% biogenic	90% biogenic	90% biogenic
Biogenic carbon capture ratio (t-CO ₂ /t-biomass)	—	1.55	1.55	1.55
Total CO ₂ removal capacity (Mt-CO ₂ per year)	0.98	2.15	1.17	2.46
% increase in CO ₂ removal capacity relative to reference system	0% (reference)	119%	0% (reference)	109%
Energy content of biomass fuel (25% moisture) (LHV) (GJ per tonne)	—	12.6	12.6	12.6
Fuel energy (LHV) to electrical energy 1 st law conversion efficiency	~52% (fuel = nat. gas)	44.9% (fuel = biomass)	24.5% (fuel = biomass)	24.5% (fuel = biomass)
Electrical energy demand (GJ per t-CO ₂) * denotes energy demand without accounting for biogenic CO ₂	~1.8	1.2 2.6*	1.0	1.3 2.4*
Electrical energy demand (MW)	~55.8	84.6	39.6	109.5
Thermal energy demand (GJ per t-CO ₂) * denotes energy demand without accounting for biogenic CO ₂	~5.6	2.8 6.1*	0.71	2.8 5.4*
Thermal energy demand (GJ h ⁻¹)	~624	727	105	886
Pressure of compressed CO ₂ (bar)	151	151	151	151
Purity of CO ₂ [% CO ₂]	97%	97%	97%	97%

n_{x,CO_2} = moles of CO₂ in stream x , n_x = moles of nonCO₂ component in stream x , X_{x,CO_2} = mole fraction of CO₂ in stream x , X_x = mole fraction of nonCO₂ components in stream x , r = residual stream, p = product stream, i = inlet stream.

The process flow diagrams shown in Fig. 2, as well as Fig. S2–S5,† include major unit operations only. Each operation contains process equipment not shown, such as pumps, compressors, mixers, and reactors. Detailed diagrams including such equipment can be found in the published techno-economic analyses that are used for process scaling (Fig. S2–S5†).^{6,12,19,20} The biomass gasification and two solid sorbent systems do not undergo intensive engineering optimization, which could further decrease costs and improve performance. We expect future studies to investigate system configurations

different than those presented here. For instance, there is potential to integrate emerging, biomass-based solid chemical looping technologies with liquid solvent DAC.^{22,23}

2.2 Economic analysis

To ensure consistency when calculating levelized costs of CO₂ removal, we follow the method of economic analysis used by Keith *et al.* for the four CO₂ removal systems.¹² Keith *et al.*'s methodology for economic analysis, which is adequate for early-stage cost estimations, requires five primary inputs: (1) itemized capital costs, (2) energy and non-energy operating costs, (3) plant utilization, (4) capital intensity, and (5) a capital recovery factor (CRF). To validate our economic model, we

independently reproduce economic figures for Carbon Engineering's baseline liquid solvent DAC process. The levelized costs calculated by the independent economic analysis match Keith *et al.*'s published values, as shown in Table S4.† Itemized capital and operating costs along with eqn (2)–(10) allow for reasonable estimates of levelized costs of CO₂ removal by the four systems.

$$\text{Levelized cost of CO}_2 \text{ removal} = \text{levelized capital cost} + \text{net operating cost} \quad (2)$$

$$= \frac{\text{Net operating cost}}{\text{CO}_2 \text{ capacity}} \quad (3)$$

$$\text{Net operating cost} = \text{nonenergy cost} + \text{energy cost} - \text{byproduct revenue}$$

$$\text{Levelized capital cost} = \text{capital intensity} \times \frac{\text{capital recovery factor}}{\text{utilization}} \quad (4)$$

$$\text{Capital intensity} = \frac{\text{total capital cost}}{\text{CO}_2 \text{ capacity}} \quad (5)$$

$$\text{Capital recovery factor} = \frac{i(i+1)^N}{(1+i)^N - 1} \quad (6)$$

$$\begin{aligned} \text{Weighted average cost of capital} &= i \\ &= (\text{interest on debt capital}) \times (\% \text{ debt financing}) \\ &+ (\text{return on equity capital}) \times (\% \text{ equity financing}) \end{aligned} \quad (7)$$

$$N = \text{project life (years)} \quad (8)$$

$$\text{Total capital cost} = \text{direct capital} + \text{indirect capital} \quad (9)$$

$$\begin{aligned} \text{Direct capital cost} &= \sum_i \text{reference cost}_i \\ &\times \left(\frac{\text{new capacity}_i}{\text{reference capacity}_i} \right)^{\text{scale factor}} \end{aligned} \quad (10)$$

We perform a sensitivity analysis on the levelized cost of CO₂ removal by varying CRF values and biomass costs. CRF is a metric that considers financing details including weighted average cost of capital (*i*) and project life (*N*) (eqn (6)–(8)). Table S2† provides CRF values with varying *i* and *N* values. Keith *et al.*'s economic analysis used CRF values of 7.5% and 12.5%, whereas this study uses values ranging from 2.5% to 17.5%. CRF values above 12.5% are reflective of stringent financial obligations that might be used for nascent CO₂ removal technologies. A CRF value of 2.5% requires a very low cost of capital, which might be possible with strong government support. Several financing uncertainties are therefore taken into account by varying the CRF.¹² As in Keith *et al.*'s economic analysis, corporate taxes are not included in this method of early-stage cost estimation. The biomass costs used for analysis range from \$20–100 per short ton, a range that includes the majority of potential biomass resources characterized in the US DOE's Billion-Ton Report.²⁴ Conservative contingency costs are used in the economic analyses of the new systems, as shown in Table 2.

The capital contingency cost for the NG + LS-DAC system is for an *n*th-of-a-kind plant with a relatively low level of uncertainty, whereas the contingency costs for the bioenergy-integrated systems are for early-stage plants with higher levels of uncertainty. To remain conservative in our economic assessments, contingencies of 20% and 30% are used for the BG + LS-DAC and BB + SS-CC/DAC systems, respectively; these contingencies were conservatively selected based on industry practices.²⁵ The NG + LS-DAC system has undergone intensive development and is currently being commercialized,¹² whereas the bioenergy-integrated systems are less robust and therefore require higher contingencies. The bioenergy-integrated solid sorbent systems have a higher contingency cost than the liquid solvent system because of the increased uncertainty in the heat exchange subsystem that is used to generate and distribute clean, low temperature steam for amine desorption, as well as the overall technical immaturity of solid sorbent DAC systems.

To account for upstream emissions, additional analyses are conducted in which emission penalties are applied to natural

Table 2 Baseline economic details for the four CO₂ removal systems analyzed in this study. The parameters for the natural gas plus liquid solvent (NG + LS-DAC) system were taken from Keith *et al.*'s economic analysis of Carbon Engineering's baseline direct air capture technology¹²

	NG + LS-DAC	BG + LS-DAC	BB + SS-CC	BB + SS-DAC
Plant utilization	90%	90%	90%	90%
Project year	2016	2016	2016	2016
Natural gas price (\$ per GJ)	3.5	—	—	—
Capital scaling factor	—	0.8	0.8	0.8
Indirect capital costs (% of total direct capital)	35%	35%	35%	35%
Capital cost contingency (% of total direct & indirect capital)	12.5%	20%	30%	30%
Excess electricity sold to grid (MW)	—	—	70	—
Purchase cost of byproduct electricity (\$ per kWh)	—	—	\$0.05	—
Excess thermal energy sold as byproduct steam (GJ h ^{−1})	—	—	780	—
Purchase cost of byproduct steam (~230 °C)(\$ per t-steam)	—	—	\$7.02	—

gas and biomass fuels.¹ The techno-economic results presented here are conservative, early-stage estimations with some uncertainty; however, they allow direct comparison between standalone and biomass-integrated DAC technologies.

2.3 Upstream emissions accounting

Baseline scenarios assume no leakage of natural gas during extraction and transportation and carbon neutrality for cultivation, harvest, and transportation of biomass. Penalty scenarios are constructed to account for upstream emissions (Tables 5 and S14†). The Penalty scenario for the NG + LS-DAC system assumes the quantity of natural gas-carbon leaked during extraction and transportation is equal to 3% of the natural gas-carbon captured on-site. A global warming potential of 25 is assigned to leaked natural gas. The penalty scenarios for the three biomass systems assume the quantity of carbon emitted to the atmosphere during cultivation, harvest, and transportation of biomass is equal to 15% of the biogenic carbon captured on-site.²⁶ Severe no bio penalty scenarios for the BG + LS-DAC and BB + SS-DAC systems are constructed in which none of the biogenic carbon captured on-site is included in the net removal of CO₂. A detailed life cycle assessment including the global warming impact of the four cases presented would provide a more accurate accounting of emissions and should be considered in future studies.

2.4 Geospatial analysis

To understand the potential for gigaton-scale CO₂ removal by the proposed BECCS–DAC systems in the US, we conduct a geospatial analysis based on methods developed by Baik *et al.*⁸ Our geospatial analysis considers the amount of biomass economically available for under \$100 per dry short ton in every US county, the locations of suitable geologic formations for permanent CO₂ sequestration, and the CO₂ removal capacity of the BECCS–DAC systems.

3. System descriptions

3.1 Natural gas plus liquid solvent direct air capture (NG + LS-DAC)

The natural gas plus liquid solvent direct air capture system (NG + LS-DAC, Fig. 2A) is adapted from Carbon Engineering's baseline liquid solvent technology that uses alkaline chemistry for CO₂ capture.¹² Carbon Engineering's baseline technology uses natural gas as the sole energy source for both thermal and electrical energy. The itemized capital and operating costs are presented by Keith *et al.* This system is analyzed and compared with the BG + LS-DAC system.

3.2 Biomass gasification plus liquid solvent direct air capture (BG + LS-DAC)

The biomass gasification plus liquid solvent direct air capture system (BG + LS-DAC, Fig. 2B) integrates biomass gasification with the NG + LS-DAC system by substituting natural gas with biomass-derived syngas. The biomass gasification technology is modeled using robust process information obtained from

a techno-economic analysis (TEA) published by the US National Renewable Energy Laboratory (NREL).¹⁹ NREL is a leading institute in the field of biomass conversion technology development and has published detailed TEAs of multiple biomass conversion technologies. The biomass gasification technology involves a high temperature, entrained-flow, slagging gasifier, which has been commercially proven by GE, Siemens, Shell, and ConocoPhillips. Corn stover is used as the model biomass feedstock with 25% moisture and an energy content of 12.6 GJ per tonne (16.8 GJ per dry tonne) (LHV); corn stover's mass composition (47% C, 5% H, 41% O, 0.2% S, 0.8% N, and 6% ash) serves as an adequate representation of lignocellulosic biomass. The original design by NREL involves biomass gasification plus catalytic conversion of syngas to liquid fuels. Extraneous operations involved with liquid fuel synthesis are not included in the BG + LS-DAC model; NREL's detailed mass and energy flows for each unit operation allow for complete system integration with a liquid solvent DAC technology. The syngas is cleaned *via* scrubbing of particulates, ammonia, hydrogen sulfide, and some CO₂. Syngas has been shown to be a suitable replacement to natural gas as fuel for gas turbines.^{19,27,28} A complete water–gas shift reaction could be incorporated to remove CO₂ pre-combustion and generate a pure stream of hydrogen instead of syngas,⁴ however, we did not model such a configuration. The costs associated with removing CO₂ from turbogenerator flue gas were deemed acceptable because of the large quantity of liquid solvent already on-site for CO₂ removal from air; Carbon Engineering's baseline design incorporates post-combustion removal of CO₂ from turbogenerator flue gas. The biomass feed rate of 2300 dry tonnes per day provides energy to remove 0.98 Mt-CO₂ per year from air, which is equivalent to that of the NG + LS-DAC system. The syngas turbogenerator and heat recovery steam generator supply enough heat (727 GJ h^{−1}) and power (84.6 MW) to operate the entire BG + LS-DAC system. The 1st law efficiency of converting biomass energy (LHV) to electricity is 44.9%, accounting for the gas turbogenerator and heat recovery steam generator. Electrical energy produced from the steam slaker excess heat is conservatively taken to be zero. The BG + LS-DAC system captures more total CO₂ than the NG + LS-DAC system due to the relatively high carbon/hydrogen ratio of biomass. Similar to coal, biomass produces more CO₂ than natural gas per unit energy harnessed *via* combustion.²¹ The CO₂ concentrations of air and syngas combustion flue gas are 0.04% and 11% by volume, respectively. The percentages of CO₂ captured from air and flue gas are 74.5% and 90%, respectively, and the end purity is 97%, equaling those in the NG + LS-DAC system. Therefore, according to eqn (1), the theoretical minimum energy inputs required for separation of CO₂ from air and flue gas are 20.38 kJ mol^{−1} (463 MJ per t-CO₂) at 25 °C and 8.34 kJ mol^{−1} (190 MJ per t-CO₂) at 90 °C, respectively (not including compression). The real work required for CO₂ separation by the BG + LS-DAC system is determined by scaling heat and power demands of the NG + LS-DAC system. The relatively large quantity of CO₂ produced from the BG + LS-DAC system requires more absolute thermal and electrical energy than the NG + LS-DAC system, particularly for calcination and

compression. However, the increased net removal of CO₂ by the BG + LS-DAC system results in lower thermal and electrical energy demands per t-CO₂ removed, when compared to the NG + LS-DAC system; see Table 1 for further details. Approximate mass and energy flows are provided in Tables 1, S5 and Fig. S3.†

The itemized capital costs are scaled using appropriate scaling factors, as shown in Table S6.† Biomass cost is the sole contributor to energy operating cost with feedstock prices varying from \$20–100 per short ton. The non-energy operating costs are comprised of those associated with biomass gasification and liquid solvent CO₂ removal.^{12,19} The non-energy operating costs associated with the gasification subsystem are comprised of steam, cooling water, other raw materials, waste disposal, and fixed costs. Fixed costs for the gasification subsystem include salary, overhead (60% of total salaries), and maintenance (2% of total installed equipment cost). The non-energy operating costs associated with liquid solvent CO₂ removal include make-up streams and fixed costs (including raw materials, maintenance, labor, and waste disposal).

3.3 Biomass boiler plus solid sorbent direct air capture (BB + SS-DAC)

The biomass boiler plus solid sorbent direct air capture system (BB + SS-DAC, Fig. 2D) integrates a proven biomass boiler technology with an emerging solid sorbent amine CO₂ removal technology. The system is scaled to have the same biomass feed rate as the BG + LS-DAC system. The biomass boiler technology is modeled using process information from a techno-economic analysis published by NREL.²⁰ The solid sorbent technology is modeled using technical information from the National Academy of Sciences.⁶ Emerging solid sorbent technologies are designed to capture CO₂ from gaseous streams concentrated in CO₂ (flue gas) as well as from streams dilute in CO₂ (air).^{13,14} The biomass boiler features a live-bottom grated fuel bin, combustor, boiler, and turbogenerator. The biomass boiler flue gas is cleaned *via* particulate removal and desulfurization. As with the BG + LS-DAC system, corn stover is the model biomass feedstock with 25% moisture and an energy content of 12.6 GJ per tonne (16.8 GJ per dry tonne) (LHV). The biomass boiler and turbogenerator generate enough heat (886 GJ h⁻¹) and power (109 MW) to operate the solid sorbent CO₂ removal system. All of the generated heat and power are used for direct air capture, with no excess. The 1st law conversion efficiencies of biomass energy (LHV) to heat and power are 55% and 24.5%, respectively.¹⁸ A heat exchange network converts the medium temperature steam (~230 °C) from the turbine outlet to low temperature saturated steam (~100 °C) for CO₂ desorption.⁶ The solid sorbent system works by temperature/pressure swing for adsorption/desorption with a cycle time of 16 minutes (90 cycles per day).⁶ The adsorbent material lifetime and CO₂ removal capacity are 0.5 year and 1.3 t-CO₂ per kg per year, respectively.⁶ The biomass boiler flue gas is comprised of 11% CO₂ by volume and at a temperature of 60 °C. Air is comprised of 0.04% CO₂ by volume and at a temperature of 25 °C. The amine sorbent subsystem is modeled as two components (Fig. S4†): one component removes CO₂ from air with an

efficiency of 65% and the other removes CO₂ from biomass boiler flue gas with an efficiency of 90%. The CO₂ end purity is 97%. Therefore, according to eqn (1), the theoretical minimum energy inputs required for separation of CO₂ from air and biomass boiler flue gas streams are 20.1 kJ per mol-CO₂ (458 MJ per t-CO₂) and 7.66 kJ per mol-CO₂ (174 MJ per t-CO₂), respectively (not including compression). The real work required for solid sorbent amine separation of CO₂ from air is taken to be 5900 MJ per t-CO₂ (4800 MJ per t-CO₂ from heat and 1100 MJ per t-CO₂ from electricity), correlating to a 2nd law efficiency of 7.76%;²¹ the real work values are conservatively taken from the high-energy demand scenario in ref. 6. The real work required for solid sorbent amine separation of biomass boiler flue gas is taken to be 870 MJ per t-CO₂ (708 MJ per t-CO₂ from heat and 162 MJ per t-CO₂ from electricity), correlating to a 2nd law efficiency of 20%.²¹ Approximate mass and energy flows are provided in Tables 1, S8 and Fig. S4.†

The itemized capital costs were scaled using appropriate scaling factors, as shown in Table S9.† Biomass cost is the sole contributor to energy operating cost with feedstock prices varying from \$20–100 per short ton. The non-energy operating costs are comprised of those associated with the biomass boiler and solid sorbent CO₂ removal. The non-energy operating costs associated with the biomass boiler are comprised of makeup water, other raw materials, and fixed costs. Fixed costs include salary, overhead (60% of total salaries), and maintenance (2% of total installed equipment cost). Uncertainty exists in the capital cost of the heat exchange subsystem that is used to generate clean, low temperature steam for amine desorption; the 30% contingency takes this uncertainty into account. The non-energy operating costs associated with solid sorbent CO₂ removal include adsorbent material replacement, maintenance, and fixed costs (including other raw materials and waste disposal). The adsorbent material price is \$50 per kg, and therefore adsorbent material replacement contributes greatly to the final levelized cost of CO₂ removal.⁶ Due to lack of information, the salary and overhead costs associated with the solid sorbent CO₂ removal subsystem are taken to be the same as that for the biomass boiler subsystem.

3.4 Biomass boiler plus solid sorbent carbon capture (BB + SS-CC)

The biomass boiler plus solid sorbent with carbon capture (BB + SS-CC, Fig. 2C) is the same as the BB + SS-DAC system, but without any direct air capture. An excess of 70 MW of power and 780 GJ h⁻¹ of heat in the form of steam are sold to generate byproduct revenue. The biomass boiler flue gas is the only gaseous stream input to the solid sorbent subsystem and has the same composition as the flue gas in the BB + SS-DAC system. Approximate mass and energy flows are provided in Tables 1, S11 and Fig. S5.†

The itemized capital costs were scaled using appropriate scaling factors, as shown in Table S12.† Biomass is the sole contributor to energy operating cost with feedstock prices varying from \$20–100 per short ton. The contributors to non-energy operating costs are the same as those for the BB + SS-

DAC system. An excess of 70 MW of power is sold to the grid at varying prices (Table 4) and an excess of 780 GJ h⁻¹ of heat in the form of steam is sold at \$7.02 per tonne.²⁵

4. Results & discussion

As shown in Table 3A and B, levelized costs of CO₂ removal by the BG + LS-DAC system range from \$54–187 per t-CO₂ across analyzed parameters, a significant improvement over the NG + LS-DAC system's range of \$65–268 per t-CO₂. Only air-derived CO₂ may be taken into account when calculating levelized costs of CO₂ removal by the NG + LS-DAC system, as natural gas-derived CO₂ does not contribute to a net removal of atmospheric CO₂. The BG + LS-DAC system's use of renewable biomass allows for biogenic CO₂ to contribute towards net CO₂ removal, resulting in a more cost-effective approach. In addition to improving economics, CO₂ removal capacity increases by 119% when biomass-derived syngas is used in place of natural gas, and the energy demand per t-CO₂ removed is significantly reduced.

As shown in Table 3C and D, with an electricity selling price of \$0.05 per kW per h, the levelized costs of CO₂ removal range from \$80–186 and \$106–165 per t-CO₂ for the BB + SS-CC and BB + SS-DAC systems, respectively, across analyzed parameters. The

high operating cost of replacing adsorbent material every 6 months makes the BB + SS-DAC system economically unfavorable when the costs of fuel (biomass) and capital are low; however, the integrated system still outperforms its reference standalone system (BB + SS-CC) in 45% of the analyzed scenarios. The economics of the BB + SS-DAC system become more favorable, relative to the BB + SS-CC system, when the cost of grid electricity is low, as shown in Table 4. The CO₂ removal capacity increases by 109% when DAC is incorporated into a more traditional, steam-driven BECCS system like BB + SS-CC. For both the liquid solvent and solid sorbent platforms analyzed, the levelized costs of the reference systems (Table 3A and C) are relatively more sensitive to the costs of fuel and capital, as shown by the stark color change in the heat maps across analyzed parameters.

The ever-present deployment of low-cost renewable electricity, particularly from wind and solar energy, threatens the economic viability of BECCS systems that are optimized for power distribution.^{29,30} Such BECCS systems typically produce power at a higher cost than solar or wind energy systems.³¹ For these reasons, utilizing heat from biomass combustion to operate thermal-energy intensive DAC systems may be a more attractive prospect than its use for grid power generation, depending on electricity and CO₂ price. Levelized costs of CO₂

Table 3 Heatmaps showing levelized costs of CO₂ removal (\$ per t-CO₂) by the baseline systems with variation in fuel cost and capital recovery factor (CRF)

\$54 \$268					
A NG + LS-DAC					
CRF	Natural Gas Price (\$/GJ)				
	\$1.50	\$3.50	\$5.50	\$7.50	\$9.50
2.5%	\$65	\$83	\$100	\$118	\$136
7.5%	\$109	\$127	\$145	\$162	\$180
12.5%	\$153	\$171	\$189	\$206	\$224
17.5%	\$197	\$215	\$233	\$250	\$268

\$80 \$186					
C BB + SS-CC					
CRF	Biomass Prices (\$/short ton)				
	\$20	\$40	\$60	\$80	\$100
2.5%	\$80	\$94	\$108	\$122	\$137
7.5%	\$96	\$111	\$125	\$139	\$153
12.5%	\$113	\$127	\$141	\$155	\$170
17.5%	\$129	\$144	\$158	\$172	\$186

B BG + LS-DAC					
CRF	Biomass Price (\$/short ton)				
	\$20	\$40	\$60	\$80	\$100
2.5%	\$54	\$62	\$70	\$77	\$85
7.5%	\$88	\$96	\$103	\$111	\$119
12.5%	\$122	\$130	\$137	\$145	\$153
17.5%	\$156	\$164	\$171	\$179	\$187

D BB + SS-DAC					
CRF	Biomass Prices (\$/short ton)				
	\$20	\$40	\$60	\$80	\$100
2.5%	\$106	\$113	\$120	\$127	\$134
7.5%	\$117	\$124	\$131	\$137	\$144
12.5%	\$128	\$134	\$141	\$148	\$155
17.5%	\$138	\$145	\$152	\$159	\$165

capture from the BB + SS-CC system significantly increase as electricity prices decrease, as shown in Table 4; additionally, levelized costs are much higher when the steam byproduct generates zero revenue, as shown in Table S15.[†] The interaction between BECCS and future electricity grids is uncertain and may involve complexities not discussed herein, but the prospect of using bioenergy for direct air capture is intriguing considering the critical role carbon removal will play in mitigating climate change. A conceivable future scenario could involve a coupled BECCS–DAC system that operates in a similar fashion to that of a peaking power plant: when electricity is priced high, the BECCS–DAC system shifts its use of bioenergy from DAC to generating excess electricity for sale to the grid. In such a scenario, the BECC–DAC system's rates of biomass combustion and biogenic CO₂ removal remain constant, ensuring the CO₂ removal subsystem remains largely operable and doesn't experience significant downtime.

Table 4 Levelized costs of CO₂ removal (\$ per t-CO₂) by the baseline BB + SS-CC system with variation in biomass price and electricity selling price. CRF constant at 12.5%

BB + SS-CC						
\$113 \$193						
Biomass Price (\$/short ton)	Electricity Selling Price (\$/kWh)					
	\$0.00	\$0.01	\$0.02	\$0.03	\$0.04	\$0.05
\$20	\$136	\$132	\$127	\$122	\$118	\$113
\$40	\$151	\$146	\$141	\$136	\$132	\$127
\$60	\$165	\$160	\$155	\$151	\$146	\$141
\$80	\$179	\$174	\$170	\$165	\$160	\$155
\$100	\$193	\$188	\$184	\$179	\$174	\$170

The levelized costs presented up to this point assume zero upstream net emissions from extraction and transportation of natural gas or cultivation, harvest, and transportation of biomass. We conduct sensitivity analyses where emissions penalties of varying severity are applied to both natural gas and biomass.¹ Fig. 3 shows the reduced quantities of net CO₂ removal due to the emissions penalties and Tables 5 and S14[†] show the resultant increase in levelized costs.

For the BG + LS-DAC system, the penalty and no bio scenarios increase costs by 9% and 119%, respectively, relative to baseline. For the BB + SS-DAC system, the penalty and no bio scenarios increase costs by 8% and 92%, respectively, relative to baseline. Removing all biogenic carbon from economic calculations (no bio scenario) results in levelized costs in the \$300–400 range for the more feasible parameters analyzed, a range which more closely aligns with estimated costs of DAC found in the academic literature.^{6,11,32} Removing 15% of biogenic carbon from economic calculations (penalty scenario) results in a relatively small increase in levelized costs, thereby demonstrating the ability of BECCS–DAC systems to retain cost-effectiveness without biomass carbon neutrality. Additionally, the levelized costs of CO₂ removal by biomass enhanced DAC systems are less sensitive to upstream emissions than their standalone reference systems. These sensitivity scenarios further demonstrate the potential cost benefits of using biomass as fuel for DAC operations.

Deployed at scale, BECCS–DAC systems enable gigaton-scale CO₂ removal in the US without long-distance biomass or CO₂ transport, thereby overcoming a significant barrier to near-term deployment.⁸ Fig. 4 shows the technical potential for CO₂ removal by BB + SS-DAC system in all counties of the Continental US by the year 2030 based on estimates in the US Department of Energy's 2016 Billion Ton Study.²⁴ Total CO₂ removal potential is 2.9 Gt-CO₂ and 1.5 Gt-CO₂ per year for the entire US and counties collocated with suitable geologic

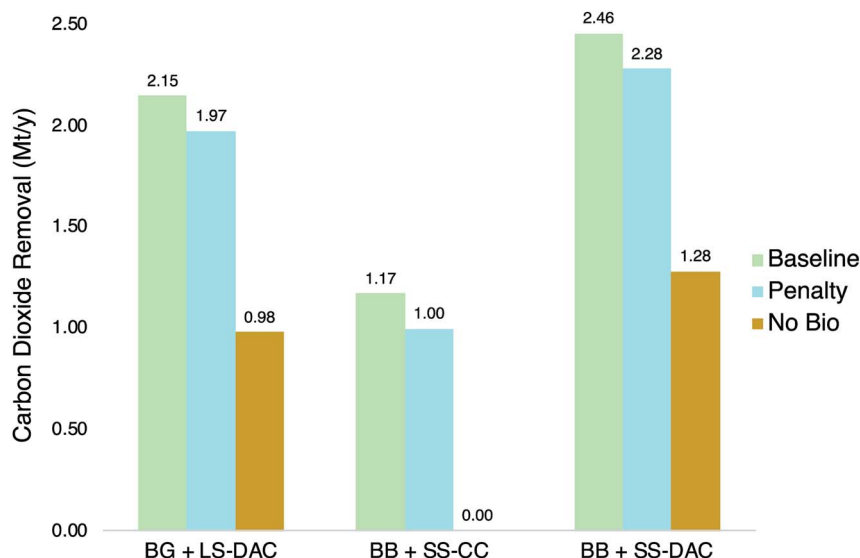


Fig. 3 Net carbon dioxide removal (Mt per year) by the biomass-enabled systems under baseline, penalty, and no bio scenarios.

Table 5 (A) Levelized costs of CO₂ removal (\$ per t-CO₂) by the BG + LS-DAC system with variation in biomass price and CRF. Baseline scenario assumes biomass is carbon neutral. Penalty scenario assumes a quantity of carbon equivalent to 15% of the biogenic carbon captured on-site is emitted to the atmosphere during biomass cultivation, harvest, and transportation. No bio scenario accounts only for air-derived CO₂ in cost calculations. (B) Levelized costs of CO₂ removal by the BB + SS-DAC system with variation in biomass price and CRF. All scenarios same as A

A																
BG + LS-DAC																
\$54 \$409																
CRF	Biomass Price (\$/short ton)															
	\$20			\$40			\$60			\$80			\$100			
	Baseline	Penalty	No Bio	Baseline	Penalty	No Bio	Baseline	Penalty	No Bio	Baseline	Penalty	No Bio	Baseline	Penalty	No Bio	
	2.5%	\$54	\$59	\$118	\$62	\$67	\$135	\$70	\$76	\$152	\$77	\$84	\$169	\$85	\$93	\$186
	7.5%	\$88	\$96	\$193	\$96	\$104	\$210	\$103	\$113	\$227	\$111	\$121	\$244	\$119	\$129	\$261
	12.5%	\$122	\$133	\$267	\$130	\$141	\$284	\$137	\$150	\$301	\$145	\$158	\$318	\$153	\$166	\$335
17.5%	\$156	\$170	\$341	\$164	\$178	\$358	\$171	\$187	\$375	\$179	\$195	\$392	\$187	\$203	\$409	
B																
BB + SS-DAC																
\$106 \$317																
CRF	Biomass Price (\$/short ton)															
	\$20			\$40			\$60			\$80			\$100			
	Baseline	Penalty	No Bio	Baseline	Penalty	No Bio	Baseline	Penalty	No Bio	Baseline	Penalty	No Bio	Baseline	Penalty	No Bio	
	2.5%	\$106	\$115	\$204	\$113	\$122	\$217	\$120	\$129	\$230	\$127	\$137	\$243	\$134	\$144	\$256
	7.5%	\$117	\$126	\$224	\$124	\$133	\$237	\$131	\$141	\$250	\$137	\$148	\$263	\$144	\$155	\$276
	12.5%	\$128	\$137	\$244	\$134	\$145	\$257	\$141	\$152	\$270	\$148	\$159	\$283	\$155	\$167	\$296
17.5%	\$138	\$149	\$265	\$145	\$156	\$278	\$152	\$163	\$291	\$159	\$171	\$304	\$165	\$178	\$317	

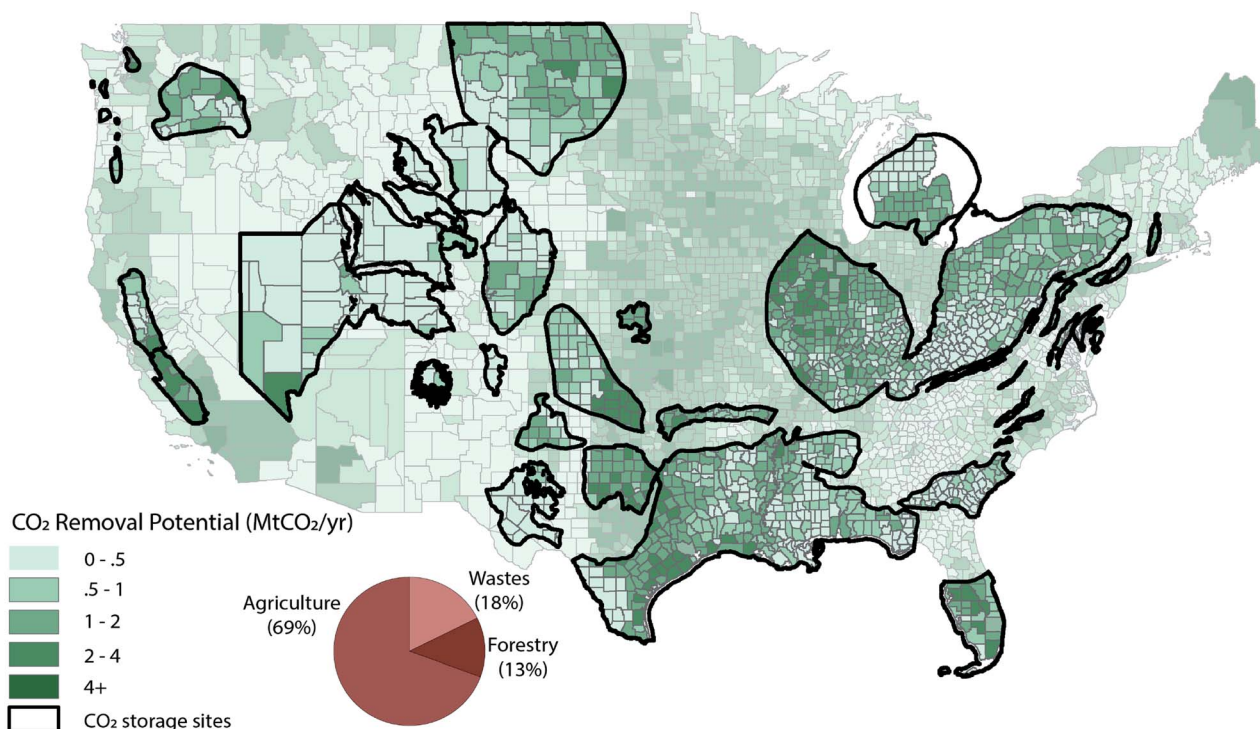


Fig. 4 County-level CO₂ removal potential of biomass integrated DAC (BB + SS-DAC) systems in the US for the year 2030 overlaid with potential geologic storage sites for CO₂. CO₂ removal potential is 2.9 Gt-CO₂ and 1.5 Gt-CO₂ per year for the entire US and counties collocated with suitable geologic formations, respectively.

formations, respectively. In comparison, the BB + SS-CC system would only remove 1.4 and 0.7 Gt-CO₂ per year, respectively, when utilizing equivalent biomass inputs.

DAC technologies have significant locational flexibility: as a result, they can easily be sited where suitable geologic storage capacity exists.³³ This locational flexibility also imparts advantages to bioenergy systems, which rely on geographically dispersed, low-energy density biomass resources. Therefore, BECCS–DAC systems extend the impact of scarce biomass resources, a primary constraint on deployment of net-zero emissions energy systems.

5. Conclusions

The coupling of BECCS and DAC technologies enables cost-effective removal of CO₂ on a gigaton scale. Biomass is a suitable energy source for DAC through its ability to provide high, medium, and low temperature thermal energy at relatively low cost while enhancing CO₂ removal capacity, as well as its inability to provide electricity at costs competitive with other renewable resources, such as wind and solar. However, DAC technologies are in their infancy and require substantial development to increase technical readiness and decrease uncertainty prior to integration with BECCS technologies. In addition, social acceptance, environmental protection, and appropriate political incentives are necessary for large-scale deployment of CO₂ removal systems fueled by biomass.^{6,9} Fortunately, the US Department of Energy has conducted thorough assessments of sustainable biomass availability to help minimize environmental and social harm as CO₂ removal systems continue to advance.²⁴ Historically, the lack of US policies placing a value on CO₂ has been a barrier to the deployment of BECCS and DAC technologies, however, the US congress recently enhanced the section 45Q carbon oxide sequestration tax credit to help overcome this barrier.³⁴ Two BECCS–DAC systems were designed and analyzed to demonstrate their economic and environmental advantages over standalone reference systems. Techno-economic analyses indicate the BECCS–DAC systems increase net CO₂ removal by 109–119% at lower costs than standalone reference systems. Moving forward, BECCS–DAC systems can drive efficient, cost-effective, and locationally-optimized usage of biomass resources for engineered CO₂ removal. As the level of attention towards CO₂ removal technology development continues to grow, BECCS–DAC systems deserve serious consideration.

Author contributions

D. L. S. and W. J. S. conceived the study; W. J. S. developed the techno-economic models with input from all coauthors; D. L. S. conducted the geospatial analysis; W. J. S. wrote the paper with input from all coauthors.

Conflicts of interest

Authors declare no conflicts of interest.

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